

FOUR STATEMENTS

Overview

Let $K \subset S^3$ be a knot, $Y_r = S_r^3(K)$, and suppose that $\phi: Y_{-2} \rightarrow Y_2$ is an orientation-preserving diffeomorphism. The Strategy computes one integer in two ways: a count Ω of instantons over a cube of metrics on a closed four-manifold $X(\Pi)$, assembled from two caps and the cobordisms of a composite Π of the maps ϕ_*B and HB . The four statements below are its four steps. In §1 (Steps 2 and 4 of the Strategy) the map B of the negative-definite cobordism $W_B: Y_2 \rightarrow Y_{-2}$, defined by an interval of metrics and cut down by a sphere of square -4 , is an integral chain map; a relation for (-4) -spheres and a constraint on the gradings of a cosmetic pair come with it. In §2 (Step 2), if B is a rational isomorphism, the Floer pairing of two caps built from the symplectic manifold of Kronheimer and Mrowka is nonzero along infinitely many composites Π_M . In §3 (Step 4) $\Omega(X(\Pi))$ equals that pairing. In §4 (Steps 1, 3, 5 and 6) $\Omega(X(\Pi_M)) = 0$ for large M , by Stokes' theorem on a cut-down moduli space of $\mathrm{SO}(3)$ -monopoles over the cube. We assume:

(H1) Floer's instanton homology of $Y_{\pm 2}$, whose only reducible flat connections are the central ones $\theta, \chi\theta$, is defined over $\mathbb{Z}$ with the absolute $\mathbb{Z}/8$ -grading fixed by θ ; a cobordism with $b_1 = 0$ carrying a class c that vanishes on its ends induces a map of degree $-2c^2 - 3b^+ \pmod{8}$ [DLME, proof of Prop. 3.4]; Donaldson invariants glue along $Y_{\pm 2}$ and along (Y_0, w) as Floer pairings; and the meridional handle cobordisms $Y_{-2} \rightarrow Y_{-1}$, $Y_1 \rightarrow Y_2$ induce isomorphisms over $\mathbb{F}_2$ (those through Y_0 do so over $\mathbb{Z}$ by Floer's exact triangle [DLME, Prop. 3.8], as $I(S^3) = 0$).

(H2) $B \otimes \mathbb{Q}$ is an isomorphism (Conjecture 1.5).

(H3) Witten's formula holds for the closed symplectic manifold X_0 of [KM04, Prop. 15]. This is [KM04, Cor. 7], whose proof rests on [FLM, Thm. 1.1], stated there under the gluing hypothesis [FLM, Hyp. 7.8.1].

(H4) Gluing in families of broken metrics at irreducible nondegenerate flat connections, and at the flat connection γ of trace zero on $L(4, 1)$ against the reducible of energy $\frac{1}{4}$ on νS , with orientations over $\mathbb{Z}$; in particular the relative sign ϵ of Theorem 1.1 is constant.

(H5) The analytic foundations of Theorem 4.1 listed in §4, uniformly in the neck length.

Main Theorem. *Let $K \subset S^3$ be a knot of genus at least two, and assume (H1)–(H5). Then there is no orientation-preserving diffeomorphism $S_{-2}^3(K) \rightarrow S_2^3(K)$.*

Proof. For the composites Π_M of Corollary 2.3, that corollary and Theorem 3.1 give $\Omega(X(\Pi_M)) \neq 0$ for infinitely many $M \in 8\mathbb{Z}$, and Theorem 4.3 gives $\Omega(X(\Pi_M)) = 0$ for all large $M \in 8\mathbb{Z}$, for the same manifolds, metrics and divisors. $\square$

By Ni–Wu [NW] and Hanselman [Ha, Thm. 2], a cosmetic pair in S^3 is $\pm 1/n$ or ± 2 , and ± 2 forces genus two; Daemi, Lidman and Miller Eismeier exclude $\pm 1/n$ [DLME, Thm. 1.3, Cor. 1.4]. So the Main Theorem would settle the cosmetic surgery conjecture in S^3 under (H1)–(H5) (VERIFIED as logic). Three corrections to the Strategy are built in: rational invertibility of B replaces the congruence $u^M \equiv 1 \pmod{2^N}$ of Step 2 (Corollary 2.3); as HB has degree four (Proposition 1.3), what is used, and expected, is only that B is an isomorphism modulo two; and the Chern–Simons filtration is not known to be favourable for B , rather than known to be unfavourable (§1).

1. The distance-four map

Let X_r° be the trace of r -surgery on K with a ball removed, and $W_B = (-X_2^\circ) \cup_J X_{-2}^\circ: Y_2 \rightarrow Y_{-2}$, $J \cong S^3$ (the W' of the Strategy). It is simply connected with form $\langle -2 \rangle^2$ on the capped Seifert surfaces F_l, F_r . The cores of the two 2-handles form a sphere S with $[S] = F_l - F_r$, $S \cdot S = -4$, $S \cap J = K$ and $\partial\nu S \cong L(4, 1)$ (VERIFIED). Let $G = [-1, 1]$ parametrize metrics broken along J at -1 and along $\partial\nu S$ at $+1$ [KM11, §3.9]. For a $\mathrm{U}(2)$ -bundle E with $\langle c_1(E), S \rangle$ even, the jumping-line divisor V_S is the set of connections A for which the holomorphic bundle $(E|_S \otimes (\det E|_S)^{-1/2}, \bar{\partial}_A)$ is $\mathcal{O}(k) \oplus \mathcal{O}(-k)$ with $k \geq 1$: the zero set of the canonical section of the determinant line of $\bar{\partial}_A$ on its twist by $\mathcal{O}(-1)$, of first Chern class $\pm\mu(S)$ [D]. On $Y_{\pm 2}$, χ is the flat real line bundle of holonomy -1 , and χ_* the involution it induces on Floer complexes (for χ -invariant perturbations, PLAUSIBLE); it has degree 4 and shifts the Chern–Simons functional by $\frac{1}{2}$, so it fixes no generator (VERIFIED).

At a fixed metric, cutting the map of W_B down by V_S gives a chain map of degree -2 . It is null-homotopic in two ways: by stretching J , since the only flat connection on S^3 is trivial, with a three-dimensional stabilizer; and, for the signed sum over the bundles with $c_1 = c$ and $c + \mathrm{PD}(S)$, by stretching $L(4, 1)$, since the two bundles agree off νS and the reducible of energy $\frac{1}{4}$ on νS enters both with opposite signs. The difference of the two null-homotopies is a chain map:

Theorem 1.1. *Assume (H1) and (H4). For $c \in H^2(W_B; \mathbb{Z})$ vanishing on $Y_{\pm 2}$, let $m_{G,c}: C(Y_2; \mathbb{Z}) \rightarrow C(Y_{-2}; \mathbb{Z})$ count the pairs $(t, [A])$, $t \in \mathrm{int} G$, with A a g_t -instanton of index one and irreducible limits on the bundle E_c with $c_1(E_c) = c$, modulo determinant-one gauge, and $[A] \in V_S$. There is a sign ϵ , depending only on orientation conventions, such that $B_c = m_{G,c} - \epsilon m_{G,c+\mathrm{PD}(S)}$ is a chain map of degree $-1 - 2c^2 \pmod{8}$, whose chain homotopy class does not depend*

on the choices. Moreover $B_{c+2a} \simeq \pm \chi_*^{a(F_r)} B_c \chi_*^{a(F_l)}$; so up to sign and homotopy there are exactly two such maps, $B = B_0$ of degree -1 and $B' = B_{-\text{PD}(F_l)} \simeq \pm B \chi_*$ of degree 3 .

Sketch. For $\dim G = 1$ the boundary formula of [KM11, §3.9] reads $\partial m_{G,c} + m_{G,c} \partial = \pm(m_{1,c} - m_{-1,c})$, where $m_{\pm 1,c}$ counts index-two solutions in V_S for the broken metric at ± 1 ; bubbles and breaking at θ , $\chi\theta$ are excluded by index. By index additivity [DLME, proof of Prop. 3.4], a solution broken along J with irreducible halves has index $\text{ind } A_l + 3 + \text{ind } A_r \geq 3$, and a reducible half has index at least -3 [DLME, Lemma 5.6] and forces a further central matching on $Y_{\pm 2}$; so $m_{-1,c} = 0$. On νS , with $\xi(S) = 1$, the reducibles with $w_2 = 0$ have $v = 2j\xi$, energy $j^2/4$ and framed index $8E = 2j^2$; those with $j = 0$ miss V_S , as they restrict to $\mathcal{O} \oplus \mathcal{O}$. As the index is $2 = \text{ind}(\text{exterior}) + 8E$, the solutions at $+1$ are rigid irreducible instantons on $W_B \setminus \nu S$ with limit γ , glued to the reducible with $j = \pm 1$; it meets V_S once, with local degree ± 1 , for its decaying deformations are the extensions of $\mathcal{O}(1)$ by $\mathcal{O}(-1)$, trivial on S unless split. Hence $m_{1,c} = \epsilon_c m_o$, where m_o counts those instantons and does not depend on c , as $\text{PD}(S)$ vanishes off νS ; and (H4) gives $\epsilon_{c+\text{PD}(S)} = \epsilon_c$. Every c is some $2a$, and $E_{c+2a} \cong E_c \otimes L_a$, L_a flat near the ends with holonomy $\chi^{a(F_l)}, \chi^{a(F_r)}$. The degree is $-2c^2 - (\text{codim } V_S - \dim G)$. $\square$

In the inequality of [DLME, Lemma 2.7(b)], B has $L - D/8 = \frac{1}{8} - \eta$, η the least energy of a counted instanton [DLME, Prop. 3.4]; this is favourable only if $\eta > \frac{1}{8}$, which is not known (VERIFIED), and is why monopoles replace the Chern–Simons filtration. The closed counterpart of Theorem 1.1 continues the relations of Kotschick, Ruberman and Fintushel–Stern for spheres of square $-1, -2, -3$ [FS1, Lemma 2.3(6), Thms. 2.1, 2.4]:

Proposition 1.2. *Let X be closed, oriented and simply connected with $b^+(X) \geq 2$, let $S \subset X$ be an embedded sphere with $S \cdot S = -4$, and $w \in H^2(X; \mathbb{Z})$ with $w \cdot S$ even. Then $D_X^{w+\text{PD}(S)}(Sz) = -D_X^w(Sz)$ for every z in the algebra generated by the point class and $S^\perp$.*

Proof for simple type. Write $\mathbf{D}_X^w = e^{Qx/2} \sum_\beta (-1)^{(w^2+\beta \cdot w)/2} a_\beta e^\beta$, the sum over the basic classes [FS2, Thm. 4.1]. As $(w + \text{PD } S)^2 = w^2 + 2w \cdot S - 4$ and $w \cdot S$ is even, replacing w by $w + \text{PD}(S)$ multiplies the term of β by $(-1)^{\beta \cdot S/2}$, and differentiating along S at $h \in S^\perp$ multiplies it by $\beta \cdot S$. So $D^{w+\text{PD}(S)}(S \cdot) + D^w(S \cdot)$ involves only classes with $\beta \cdot S \equiv 0 \pmod{4}$, $\beta \cdot S \neq 0$; by [FS2, Thm. 4.3] these have $\beta \cdot S = \pm 4$, and $\sum_{\beta \cdot S=4} a_\beta e^{\beta+S} = (-1)^{(1+b^+)/2} \sum_{\beta \cdot S=4} a_\beta e^{-\beta-S}$. With $a_{-\beta} = (-1)^{(1+b^+)/2} a_\beta$, which follows from the parity of the degrees in which D^w is nonzero, this says that the classes with $\beta \cdot S = -4$ are the $\beta + 2\text{PD}(S)$ with $\beta \cdot S = 4$, with $a_{\beta+2\text{PD}(S)} = a_\beta$. Such a pair has the same sign and the same restriction to $S^\perp$, and opposite $\beta \cdot S$, so its terms cancel. $\square$

Proposition 1.3. *Let $W_H: Y_{-2} \rightarrow Y_2$ be the composite of the four meridional handle cobordisms through Y_{-1}, Y_0, Y_1 . Every class c on W_H that vanishes on $Y_{\pm 2}$ and is odd on the capped Seifert surface $F_0 \subset Y_0$ has $c^2 \equiv 0$ or $2 \pmod{4}$, so $(W_H, c)_*$ has degree -3 or $+1$, the two cases differing by χ_* at one end. We fix c_H with $c_H^2 = 0$ and put $H = (W_H, c_H)_*$. Then HB has degree 4 , and is not chain homotopic to the identity if $I(Y_2) \neq 0$.*

Proof. $H_2(W_H)$ has a basis R_1, R_2, R_3, F_0 with $R_i^2 = -2$, $R_1 \cdot R_2 = R_2 \cdot R_3 = 1$, $F_0 \cdot R_2 = -1$ and the other products zero. The class $c = \text{PD}(x_1 R_1 + x_2 R_2 + x_3 R_3 + d F_0)$ is odd on F_0 iff x_2 is odd, and then $c^2 \equiv 2(1+d) \pmod{4}$. $\square$

Corollary 1.4. *Assume (H1), and let $\mathbb{F} = \mathbb{F}_2$ or $\mathbb{Q}$. If a cobordism $V: Y_{-2} \rightarrow Y_2$ with $b_1(V) = b^+(V) = 0$ induces an isomorphism on $I(\cdot; \mathbb{F})$, for instance the mapping cylinder of ϕ , then $\dim I_j(Y_2; \mathbb{F})$ does not depend on $j \in \mathbb{Z}/8$. In particular $\dim I(Y_1; \mathbb{F}) \equiv 0 \pmod{8}$ and $\Delta'_K(1) = 0$.*

Proof. By (H1), V_* has even degree $-2c^2$ and H odd degree, so $H \circ V_*^{-1}$ is an automorphism of odd degree, and odd numbers generate $\mathbb{Z}/8$. The isomorphism $I(Y_1) \cong I(Y_2)$ of (H1) has even degree, $\chi(I(Y_1)) = \pm 2\lambda(Y_1)$ [T1], and $2\lambda(Y_1) = \Delta''_K(1)$ by Casson’s surgery formula. $\square$

Conjecture 1.5. *Over $\mathbb{F}_2$, $B \simeq g_- \vartheta g_+$, where $g_+: C(Y_2) \rightarrow C(Y_0, w)$, $g_-: C(Y_0, w) \rightarrow C(Y_{-2})$ are the distance-two maps of [DLME, Prop. 3.10] for the triads (Y_2, S^3, Y_0) , (Y_0, S^3, Y_{-2}) , both isomorphisms, and ϑ is the identification at Y_0 fixed by a family of metrics on $W_B \# (S^2 \times S^2)$. So $B \otimes \mathbb{F}_2$ and $B \otimes \mathbb{Q}$ are isomorphisms.*

Status. VERIFIED: the index counts behind Theorem 1.1, including the framed index $8E$ of the reducibles on νS (close νS up in $\mathbb{CP}^2$), and its twisting and degrees; Proposition 1.2 for simple type, given [FS2, Thms. 4.1, 4.3], and on $K3 \# 4\mathbb{CP}^2$ and $E(4) \# 4\mathbb{CP}^2$ by computer; Proposition 1.3; Corollary 1.4 given (H1), which recovers the condition $\Delta'_K(1) = 0$ of [BL]; the step from $\mathbb{F}_2$ to $\mathbb{Q}$ in Conjecture 1.5 (the cone of B then has finite homology of odd order). PLAUSIBLE: the chain property, i.e. (H4) (80%); Proposition 1.2 in general, by stretching $L(4, 1)$ (75%). SPECULATIVE: that the condition modulo 8 is new. Open: Conjecture 1.5 (85% that $g_\pm$ are isomorphisms, 60% that $B \simeq g_- \vartheta g_+$). Confidence for §1: 75%.

2. Nonvanishing

Lemma 2.1. *Let A be an automorphism of an r -dimensional rational vector space, and ψ_-, ψ_+ a vector and a covector with $\psi_+(\psi_-) \neq 0$. For every $k \geq 1$ the sequence $\psi_+(A^{kM} \psi_-)$, $M \geq 0$, has no r consecutive zeros.*

Proof. The characteristic polynomial of A^k has nonzero constant term, so the sequence satisfies a recurrence of order r that can be solved backwards; r consecutive zeros would propagate to $M = 0$. $\square$

Let $g(K) \geq 2$. Then Y_0 carries a taut foliation with the capped Seifert surface $\hat{\Sigma}$ of genus $g(K)$ as a leaf [Ga], and $X_0 = X^- \cup_{Y_0} X^+$ is the closed symplectic manifold of [KM04, Prop. 15]: $H_1(X_0) = H_1(X^\pm) = 0$, $b^+(X^\pm) > 0$, and there is w with $\langle w, \hat{\Sigma} \rangle = 1$. Let $f_+ : I(Y_0, w) \rightarrow I(Y_2)$ and $f_- : I(Y_{-2}) \rightarrow I(Y_0, w)$ be the maps of the cobordisms $W_{0 \rightarrow 2}$, $W_{-2 \rightarrow 0}$ of two meridional 2-handles each, so that $H = f_+ f_-$, and put $C_- = X^- \cup W_{0 \rightarrow 2} : \emptyset \rightarrow Y_2$ and $C_+ = W_{-2 \rightarrow 0} \cup X^+ : Y_{-2} \rightarrow \emptyset$; they have $H_1 = 0$ and $b^+ \geq 2$ (VERIFIED).

Proposition 2.2. *Assume (H1) and (H3). There are $z_\pm$ in the algebras generated by the point class and $H_2(X^\pm)$ with $\langle \Psi_{C_+}(z_+), H^{-1} \Psi_{C_-}(z_-) \rangle = \pm D_{X_0}^w(z_- z_+) \neq 0$ in $\mathbb{Q}$.*

Proof. Gluing along (Y_0, w) gives $\Psi_{C_-} = f_+ \Psi_{X^-}$ and $\Psi_{C_+} = \Psi_{X^+} f_-$, hence the identity. By Mayer–Vietoris, $H_2(X^-) \oplus H_2(X^+)$ maps onto $\hat{\Sigma}^\perp$, and as $H_1(X_0) = 0$ and $\hat{\Sigma}$ is primitive, two characteristic classes restrict equally to $\hat{\Sigma}^\perp$ iff they differ by some $2t \text{PD}(\hat{\Sigma})$. By (H3) the restriction of $D_{X_0}^w$ to $\hat{\Sigma}^\perp$ is $c e^{Q/2}$, $c \neq 0$, times a combination of exponentials, one for each coset, whose coefficients are signed sums of Seiberg–Witten invariants over it. The leaf $\hat{\Sigma}$ is symplectic for the form of [KM04, Prop. 15], so the canonical class β_ω has $\beta_\omega \cdot \hat{\Sigma} = 2g - 2$ and $(\beta_\omega + 2t \text{PD} \hat{\Sigma})^2 = \beta_\omega^2 + 4t(2g - 2)$. For $g \geq 2$ its Seiberg–Witten moduli space has negative dimension if $t < 0$ and positive dimension if $t > 0$, where simple type kills it. So the coset of β_ω contributes $\pm c \text{SW}(\beta_\omega) = \pm c \neq 0$ [T2]; expand multilinearly. $\square$

Corollary 2.3. *Assume (H1)–(H3), $g(K) \geq 2$ and that ϕ exists, and put $u = (\phi_* B)^3 H B$. For every p_0 there are a composite Π_0 of $\phi_* B$ and $H B$ that begins and ends with at least p_0 factors $\phi_* B$ and has at least three factors $\phi_* B$ between consecutive factors $H B$, and $p' \in 8\mathbb{Z}$ with $p' \geq p_0$, such that $\Pi_M = B \Pi_0 u^M (\phi_* B)^{p'}$ satisfies $\langle \Psi_{C_+}(z_+), \Pi_M \Psi_{C_-}(z_-) \rangle \neq 0$ for infinitely many $M \in 8\mathbb{Z}$.*

Proof. By (H1)–(H2), $H B$ and $\phi_* B$ are rational automorphisms. Cayley–Hamilton for $H B$ writes $H^{-1} = B(H B)^{-1}$ as a combination of the $B(H B)^j$, $j < r$, so by Proposition 2.2 some $B(H B)^j$ pairs nontrivially. As 1 lies in the span of the powers $(\phi_* B)^\ell$, $\ell \geq L$, for every L , inserting such combinations between consecutive factors $H B$ and at both ends gives a composite $B \Pi_0$ of the required form that pairs nontrivially. Lemma 2.1 for $\phi_* B$ and then for u , with $k = 8$, gives p' and infinitely many M . $\square$

Status. VERIFIED: Lemma 2.1; Proposition 2.2 given (H1), (H3) and Gabai’s leaf (PLAUSIBLE, standard), the topology of X_0 and of the caps having been checked against [KM04, Prop. 15]; Corollary 2.3 as a deduction. Nothing is known about the parity of $D_{X_0}^w(z_- z_+)$, so the nonvanishing must be carried over $\mathbb{Q}$, as here; integrality matters only because Theorem 4.1 yields $2^{n_a-1} \Omega = 0$, which is empty modulo two once $n_a \geq 2$ (VERIFIED). PLAUSIBLE: (H3) (85–90%, conditional on [FLM, Hyp. 7.8.1]). Open: (H2) (about 65%). The genus enters the argument only here, through $\beta_\omega \cdot \hat{\Sigma} = 2g - 2$. Confidence: 95% for the deductions, 85% for Proposition 2.2 with (H3) as a theorem.

3. The composition law

Let Π be a composite of B , H and maps induced by orientation-preserving diffeomorphisms between copies of $Y_{\pm 2}$, and $X = X(\Pi)$ the closed manifold obtained by gluing the corresponding copies of W_B and W_H along those diffeomorphisms and capping with caps $C_\pm$ of the two ends, with $H_1 = 0$, $b^+ \geq 1$ and fixed metrics, carrying classes $z_\pm$ (as in §2). Let J_i , S_i ($1 \leq i \leq n$) be the three-spheres and spheres in the copies of W_B . Let c_0 restrict to the classes of the caps, to c_H and to 0, and let E_e , $e \in \{0, 1\}^n$, have $c_1(E_e) = c_0 + \sum_i e_i \text{PD}(S_i)$; these bundles have the same κ and pairwise distinct w_2 (VERIFIED). Let g_t^R , $t \in Q = [-1, 1]^n$, restrict on the i -th copy of W_B to the interval G of §1 in t_i , with the necks along the gluing three-manifolds of length R , except the two inside each copy of $W_B \cup W_H \cup W_B$, which have fixed length and metric. With $\epsilon(e) = \prod_i (-\epsilon)^{e_i}$, put

$$\Omega(X; g^R) = \sum_e \epsilon(e) \# \{ (t, [A]) : t \in \text{int } Q, A \text{ a } g_t^R\text{-instanton on } E_e, [A] \in V_{S_1} \cap \cdots \cap V_{S_n} \cap \mathcal{V}(z_-) \cap \mathcal{V}(z_+) \}.$$

Each J_i separates X into two parts with $b^+ \geq 1$, so all Donaldson and Seiberg–Witten invariants of X vanish [D]; at a fixed metric the problem cut down by $\mu(S_1) \cdots \mu(S_n)$ has virtual dimension $-n$, and the faces of the cube contribute nothing in total, for the two reasons of §1.

Theorem 3.1. *Assume (H1), (H4), and that consecutive copies of W_H are separated by at least three copies of W_B . For generic auxiliary data in which the moduli spaces of irreducible instantons on the two halves of each W_B are transverse to the jumping-line divisors of the capped halves of S_i (Lemma 3.2), there is R_0 such that for $R \geq R_0$ the count $\Omega(X; g^R)$ is finite, independent of R and of the data away from the caps, and $\Omega(X; g^R) = \pm \langle \Psi_{C_+}(z_+), \Pi \Psi_{C_-}(z_-) \rangle$. Both sides vanish unless $c_2(E_0) \in \mathbb{Z}$. If all necks are long, the condition on the copies of W_H is not needed.*

Lemma 3.2. *Let $[A_\nu] \in V_{S_i}$ for metrics with $t_i \rightarrow -1$ converge to (A_l, A_r) , matched at the trivial connection on J_i , with no concentration on S_i . Then A_l or A_r lies in the jumping-line divisor of its capped half of S_i .*

Proof. On the nodal curve the twist $\mathcal{O}(-1)$ becomes $\mathcal{O}(-1)$ on one component and $\mathcal{O}$ on the other, and the limiting index-zero operator has kernel $\{(\sigma_l, \sigma_r) \in H^0(E_l(-1)) \oplus H^0(E_r) : \sigma_l(p) = g\sigma_r(p)\}$, which vanishes, for every gluing element g , iff $E_l \cong E_r \cong \mathcal{O} \oplus \mathcal{O}$. Invertibility passes to large ν by linear gluing of Cauchy–Riemann operators. $\square$

Sketch of proof of Theorem 3.1. Pass to ideal limits as $R \rightarrow \infty$ [KM11, §3.9]. Index additivity gives $d = \sum_P d_P + \sum_\tau \text{ind } \tau + 3c$: d_P is the expected dimension of the cut-down problem on the piece P , with its parameters, τ ranges over the trajectories on the long necks, and c is the number of matchings at central flat connections there. A reducible on a copy of W_B lying in V_S is even on both halves, so it has energy at least $\frac{1}{2}$ and index $8E - 3 \geq 1$ (Proposition 4.2); one with $\langle v, S \rangle = 0$ meets V_S only with a point of concentration on S ; and by Lemma 3.2 a copy split along J_i carries the constraint on one half. With the transversality assumed, $d_P \geq 0$ for every piece, except that a copy of $W_B \cup W_H \cup W_B$ with fixed internal necks has $d_P \geq -1$ with one central limit and $d_P \geq -4$ with two. As copies of W_H are three copies of W_B apart, a central neck borders at most one such piece, and $d \geq 2$ once $c > 0$. Hence in dimension zero $c = 0$, and the limits are rigid irreducible solutions on the pieces, matched at irreducible flat connections; each glues uniquely by (H4). The weights factor, which gives the identity at chain level. Invariance follows from it, as each piece’s map is invariant up to homotopy: B by Theorem 1.1, and H because a reducible on a wall of W_H , with its two central matchings, raises the index by at least 5. $\square$

Status. VERIFIED: the index arithmetic, by an exhaustive computation over the possible limits, and the need for each hypothesis. Without Lemma 3.2, b copies of W_B split along J with trivial halves reach expected dimension $3 - b$; with two copies of W_B between copies of W_H , k adjacent copies of $W_B \cup W_H \cup W_B$ reach $3 - k$; without the transversality, a copy of $W_B \cup W_H \cup W_B$ with two central limits, between copies of W_B split along J with reducible halves facing it, reaches 0 (with all necks long, dimension zero needs no transversality). Also VERIFIED: the algebra of Lemma 3.2; the distinct w_2 . PLAUSIBLE: compactness and gluing for families of broken metrics whose cuts carry reducibles, excluded in [KM11, §3.9] and handled by index bounds in [DLME, §5]; the linear gluing in Lemma 3.2; the transversality, a genericity statement; the integral sign (85–90%). Open, and not needed: invariance when the long necks are shortened. Confidence: 80% with all necks long, 70% as stated.

4. Vanishing in families

Let X be closed and oriented with $H_1(X; \mathbb{Z}) = 0$ and $b^+ \geq 1$, containing disjoint separating three-spheres J_i and disjoint spheres S_i ($1 \leq i \leq n$) with $S_i \cdot S_i = -4$, $S_i \cap J_i$ a knot and $S_i \cap J_j = \emptyset$ for $j \neq i$; $[S_i] = F_{l,i} - F_{r,i}$, halves of square -2 . Let $(g_t)_{t \in \bar{Q}}$ stretch J_i as $t_i \rightarrow -1$, and $\partial \nu S_i$, with a metric of positive scalar curvature, as $t_i \rightarrow +1$. In the notation of [FL2b], let $\mathfrak{t}_e$ be spin ^{u} structures with $w_2(\mathfrak{t}_e) \equiv w_0 + \sum_i e_i \text{PD}[S_i]$, $\Lambda_e = c_1(\mathfrak{t}_e)$ and common $p_1 = -4\kappa$, where w_0 is even on the halves and odd on a class orthogonal to the S_i , and $\langle \Lambda_0, S_i \rangle = 2$, which is exactly the condition for $n_a = \frac{1}{4}(p_1 + \Lambda_e^2 - \sigma)$ (the n_D of the Strategy) to be independent of e (VERIFIED). Let $z = \mu(S_1) \cdots \mu(S_n)z'$ with $\deg z = d_a + n$, $d_a = 8\kappa - 3(1 + b^+)$, and $\Omega = \sum_e \epsilon(e) \#(\bar{\mathcal{V}}(z) \cap \bar{M}_\kappa^{w_e}(Q))$, as in §3. Let $\mathcal{Z}_e = \mathcal{V}(z) \cap \mathcal{W}^{n_a-1} \cap \mathcal{M}_{\mathfrak{t}_e}^{*,0}(Q)/S^1$ be the cut-down one-manifold of [FL2b, (3.62)], where the sections defining $\mathcal{W}$, which represent μ_c [FL2b, (3.12)], sample the spinor on every piece. (H5) consists of: compactness of $\mathcal{Z}_e$ over $\bar{Q}$; transversality for single components, for the product problem at broken limits (modulo the gauge groups of the pieces and the diagonal circle, by multivalued perturbations) and for the problem obtained by forgetting the components that are abelian instantons; [FL2b, Prop. 3.29] with parameters; gluing at the lens faces with the sign ϵ of Theorem 1.1; and the control of Seiberg–Witten classes on the positive pieces and the caps used in Theorem 4.3.

Theorem 4.1. *Assume (H5) and $n_a \geq 1$, and that the cut-down instanton spaces defining Ω are finite sets of regular points over $\text{int } Q$. Suppose that, for every e , no point of the closure of $\mathcal{Z}_e$ over $\bar{Q}$ is reducible on one of the pieces into which the broken necks cut X , other than the νS_i ; that is, the closure meets no stratum $\iota(M_s(\hat{\Gamma})) \times \text{Sym}^\ell(\hat{\Gamma})$ of such a piece $\hat{\Gamma}$, at any level $\ell \geq 0$, its zero-spinor part included. Then $\Omega = 0$.*

Sketch. This is the proof of [FL2b, Thm. 3.33(a)] with faces. By [FL2b, Prop. 3.29] the link of the instanton stratum contributes $\pm 2^{n_a-1} \#(\bar{\mathcal{V}}(z) \cap \bar{M}_\kappa^{w_e}(Q))$ ends to $\mathcal{Z}_e$. A lower level lowers the dimension by six, and a point of concentration frees conditions of degree at most four [FL2b, (3.25)]; a broken J_i lowers it by four, three for the stabilizer of the trivial connection and one for the parameter, given the transversality of the product problem. At $t_i = +1$ only the reducible of energy $\frac{1}{4}$ on νS_i occurs; the exterior problems for e and $e + e_i$ on $X \setminus \nu S_i$ coincide, since $\text{PD}(S_i)$ vanishes there, and their ends cancel in the weighted sum. By hypothesis there are no other ends, and Stokes’ theorem gives $2^{n_a-1} \Omega = 0$ in $\mathbb{Q}$. $\square$

For $n = 0$ this is [FL2b, Thm. 3.33(a)], with its hypothesis (3.68), no reducible at any level, weakened to “none in the closure of $\mathcal{Z}$ ”; at a fixed metric with spheres it is Theorem 2.1 of the Strategy. Nothing is glued along a Seiberg–Witten stratum (VERIFIED as logic). Reducibles are excluded by energy:

Proposition 4.2. *Let the halves be pairwise orthogonal, π the orthogonal projection to the complement of their span, and $v \equiv w_0 \pmod{2}$. Then $-\frac{1}{4}v^2 \geq -\frac{1}{4}(\pi v)^2 + \frac{1}{2} \# \{i : \langle v, S_i \rangle \neq 0\}$.*

Proof. $-v^2 = -(\pi v)^2 + \frac{1}{2} \sum_i (\langle v, F_{l,i} \rangle^2 + \langle v, F_{r,i} \rangle^2)$, the evaluations are even as those of w_0 are, and one of them is nonzero if $\langle v, S_i \rangle \neq 0$. $\square$

A reducible limit of points of V_{S_i} with $\langle v, S_i \rangle = 0$ has a point of concentration on S_i ; as the S_i are disjoint, its level satisfies $\ell \geq T(v) = \#\{i : \langle v, S_i \rangle = 0\}$ (VERIFIED).

Theorem 4.3. *Let $X = X(\Pi_M)$ for the composites of Corollary 2.3, with the caps blown up once, $\langle \Lambda_0, e_{\pm} \rangle$ odd and large, and the family g^R of §3, $R \geq R_0(M)$. Assume (H5), with constants independent of R . Then for all large $M \in 8\mathbb{Z}$, $n_a \geq 1$ and the hypotheses of Theorem 4.1 hold, so $\Omega(X(\Pi_M); g^R) = 0$.*

Sketch. Blowing up does not change $\Psi_{C_{\pm}}(z_{\pm})$ [FS1], so §§2–3 apply, and Theorem 3.1 gives the finiteness. Let m be the number of copies of W_H . Each copy of W_B , with one parameter and one condition of degree two, raises 8κ by 1, and each positive piece $P = X_{-2}(K) \cup W_H \cup (-X_2(K)) \cong \mathbb{CP}^2 \# 5\overline{\mathbb{CP}^2}$ raises b^+ , hence 8κ , by 3. As $\langle \Lambda, S_i \rangle = 2$, Λ is even on the halves with difference two, so the negative pieces $X_{-2}(K) \cup_{\phi} (-X_2(K))$ add only a bounded amount to $\Theta = \frac{1}{4}(\Lambda^2 - \sigma)$, while $\Theta(P) = 1$. So $n_a = \Theta - \kappa = \frac{1}{8}(5m - n) + O(1)$. Proposition 4.2, and on P the control of Seiberg–Witten classes (a class on the indefinite P may pair nontrivially with both adjacent spheres at energy $\frac{1}{2}$ rather than 1), give $-\frac{1}{4}v^2 + T(v) \geq \frac{1}{2}(n - m) - C$; so an unbroken reducible has $\ell - T(v) \leq \frac{1}{8}(7m - 3n) + O(1)$. For Π_M , $n = 4M + O(1)$ and $m = M + O(1)$, so $n_a \rightarrow \infty$ and $\ell - T(v) \rightarrow -\infty$. On a chain of reducible components between broken necks the same inequalities give expected dimension at least -2 (attained by one positive piece between two broken necks); the count of Theorem 4.1 needs at least -3 , and -2 next to a component with nonzero spinor, granted the transversality after forgetting the abelian instantons. The caps carry no abelian instanton, as w is odd on $\hat{\Sigma}$ and their periods are generic, and $\langle \Lambda_0, e_{\pm} \rangle$ large excludes their Seiberg–Witten classes. $\square$

The same count shows why a family is needed. At a fixed metric $8\kappa = 2n + 3m + O(1)$, and $n_a \rightarrow \infty$ needs $m/n > \frac{2}{5}$ while exclusion needs $m/n < \frac{2}{7}$; over the cube the range is $\frac{1}{5} < m/n < \frac{3}{7}$, and u has $m/n = \frac{1}{4}$ (VERIFIED). Without ϕ only $(HB)^M$ is available, with $m/n = 1$; its pairings are nonzero for infinitely many M , by the argument of Corollary 2.3, and the exclusion fails, as it must (VERIFIED).

Status. VERIFIED: Theorem 4.1 as a deduction from (H5), following [FL2b, §3]; Proposition 4.2; the dimension counts and the ranges above; the energy bound on P , a lattice inequality checked by hand and exhaustively; the bounds on chains, with -2 attained; Theorem 4.3 as a deduction. PLAUSIBLE, routine: compactness, single components, the link with parameters, weighted Stokes. PLAUSIBLE, new: the transversality of the product problem and after forgetting the abelian instantons (50–55%); the lens ends, whose sign must be that of Theorem 1.1 (70%); the control of Seiberg–Witten classes on P uniformly over $\bar{Q}$, and the cap estimates (65%). The caps have $H_1 = 0$, which suffices where simple connectivity was assumed, as it enters only through the goodness of w_e [FL2b, Def. 3.20] (PLAUSIBLE). Open: a proof of the transversality after forgetting the abelian instantons (below); gluing at abelian instantons in families, which is not needed. Confidence that $\Omega(X(\Pi_M)) = 0$ for large M by this route: about 30%.

The hardest open point

It is the transversality of the problem obtained by forgetting the abelian instantons, at broken limits that contain both a component with nonzero spinor and a chain of abelian instantons. On negative pieces such instantons exist for every value of the parameters, and their own circle of stabilizers keeps a cokernel of negative index that no equivariant perturbation removes; so they can be neither made regular nor excluded by counting, and Feehan and Leness reach their closed analogue only through gluing hypotheses [FLM, Hyps. 7.8.1, 11.3.5]. The remaining problem must be made transverse by multivalued perturbations coupling distant pieces through the phase of the spinor, with a margin of one dimension. Verdict: the four statements prove the Main Theorem from (H1)–(H5) (VERIFIED as logic; 80% that no further compatibility condition is missing). (H1) and (H3) at 85–90%, (H2) at about 65% and (H4) at 80% are PLAUSIBLE; (H5), at about 30%, is SPECULATIVE as a whole. They are positively correlated, and the cosmetic conclusion by this route is SPECULATIVE, at about 15% (10–25%).

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